

SimTrans: A Simulation tool for sustainable freight transport policies

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ABSTRACT. The authors have been asked by the French Ministry of Transport, to develop a System Dynamics model which could help analyze the dynamics of the French freight transportation market (Road – Rail – River). A little later, the model was modified and used to analyze the possible consequences, on that same market, of the Kyoto proposed Emission credit policy.

Both models are explained here, and main results are shown. In addition we explain how an « ex post » causal loop analysis can help, in this context, to have experts understand better the numerous and often contradictory simulation results.

KEYWORDS: . System dynamics, transportation, greenhouse gas, transport policies, Co2, modal split, feedback, negotiable emission permits, oil, truck, road, rail, waterways.

1.Transport policies and Kyoto agreements

Transport – and particularly that of freight – is perceived socially as a vector of harmful effects and generation of greenhouse effect gas (European, 2001). The correlation between the economic growth and that of the exchanges, on the international (Ecmt, 2002) or intercontinental level as well as from a strictly national point of view (Salini *et al.*, 1981, 1982, 1983) leads naturally to try to analyze the

means of weighing on the modal split, favoring those means of transport whose external effects are least negative (TERM, 2002). It also led the European Commission to recommend policies which – beyond the sole modal split – would weigh on the development of traffic, namely on the “link” between economic growth and mobility.

These global problems and issues – whose link with the Kyoto agreements is evident – thus lead us to explore the impact on traffic and modal split, of transport policies and of the tools available to fight against the greenhouse effect. Building the SimTrans model (Salini and Karski, 2000), and later SimTransCO2 (Kbs, 2002) (the latter taking into account the introduction of negotiable emission permits), is an attempt at answering the first set of problems, without looking at the “decoupling” question, but rather searching the impact of transport policies and the fight against the greenhouse effect.

2. SimTrans : a simulation model based on System Dynamics

The approach we chose, based on System Dynamics, that is on the development of a simulator containing a large number of nonlinear relations and feedback loops, was, when we started the development, relatively original in France and Europe (Schade, 2000).

Traditionally, the models used to establish forecasts in the area of transport are based on a relatively linear logic – more or less complex – linking or combining the generation and the spatial distribution of the traffics and their modal and axial spreading. The input variables are traditionally those needed for macroeconomic positioning, as well as internal or environmental ones, specific to transport such as costs or price, investment level, etc.

The models used are generally based on families of multiple linear regressions relating to historical values of variables which can take into account in a related or a concurrent way, methods based on the analysis of so called revealed or declared “users” preferences. With regard to goods, knowledge of users behavior (Houée *et al.*, 2002) does not have the same meaning as with travellers. Strictly speaking, a travel does not have the same complex bond with other activities as it can when sending goods. Furthermore, such shipments are seldom managed independently of other shipments. Lastly, the formalization of parameters describing such mailings is extremely difficult and can need complex and questionable typologies (Think-up, 2001).

All that led to a prevalence of rather simplistic macro-economic methods of forecasting of freight transport. Only the possibility of a fine decomposition of traffic by type of product allows a real diversity of analysis of freight transport (Pretram, *op. cit* see (Salini *et al.*, 1981; 1982; 1983)). Recently (because of the

relatively short history of road transport prices), price effects, and infrastructure developments, have been added to somewhat classical forecast devices.

As far as SimTrans is concerned, it was initially decided not to break the direct connection linking macro-sectorial variables basic to the understanding of traffic generation. Rather, we wanted to take into account “the feedback loops which, through level and rate variables, delays and non-linearities, determine the dynamics of a system” (Sterman, 2000). We felt that, in a very classical way, the transport system was functioning in a world of competition between firms and transport modes, and that the behavior of a mode in response to a signal (for example a change in fuel taxation) was not limited to a change in cost, but also meant some complex reactivity of the system.

This action-reaction principle, or feedback, is one of the main market adjustment device, but it is also characteristic of adapting type behaviors. Thus for example, one can imagine that a tax rise would result in an increase of production unit cost, but would also modify the system productivity and the way of controlling it. In the longer term, it can modify investissement policies and the choice of production techniques.

This shows two well known types of cause-effect relations:

- A tax increase contributes to raise costs, which leads – somewhat indirectly – to a loss of attractivity, hence a drop in the sale of the taxed product.
- In same time, this same taxation leads to a gain in productivity...which goes against the aforementioned effect of taxes, and thus increases the attractivity of the product. On a longer term, changes of production tools (machines, vehicles) or techniques will have the same “negative loop” type effect.

These phenomena will intervene at different moments in time, with different intensities and modulations. This is an essential characteristic of System Dynamics : taking into account the interrelationships between stocks and flows or if one prefers, between resources or capacities and physical, financial or informational flows which characterize economy.

This concept of “capacity” is essential for us. Everyone knows that transport systems contain various capacity constraints:

- infrastructure constraints which investment, wear – linked to use – and techniques can modify;
- constraints of transport means – trucks, boats, trains – whose evolution is also linked to investment, obsolescence and technical standards and regulations;
- constraints linked to time, since freight cannot be stored nor much delayed.

However competition implies basically that each competitor reacts to one’s own environment, but also to reactions of other competitors. Thus, a price change, a variation of capacity, a productivity change lead to a change in competing balances in time.

Thus it is a second and fundamental characteristic of System Dynamics: rather than taking into account isolated events, it combines them, as well as their evolution, in time evolving events packs. In practice, this led us to model a transport system with three competitive ground modes, each with a certain freight transport capability, used with a certain changing productivity and variable quality.

The logic in each “modal” model is identical, the difference being in each mode’s specificity. For example, the railway capacity will refer to the concept of path (capacity available at a given moment between two points), whereas this concept is meaningless for river or road.

The following is a simplified causal diagram of the model, showing only one mode of transportation (road).

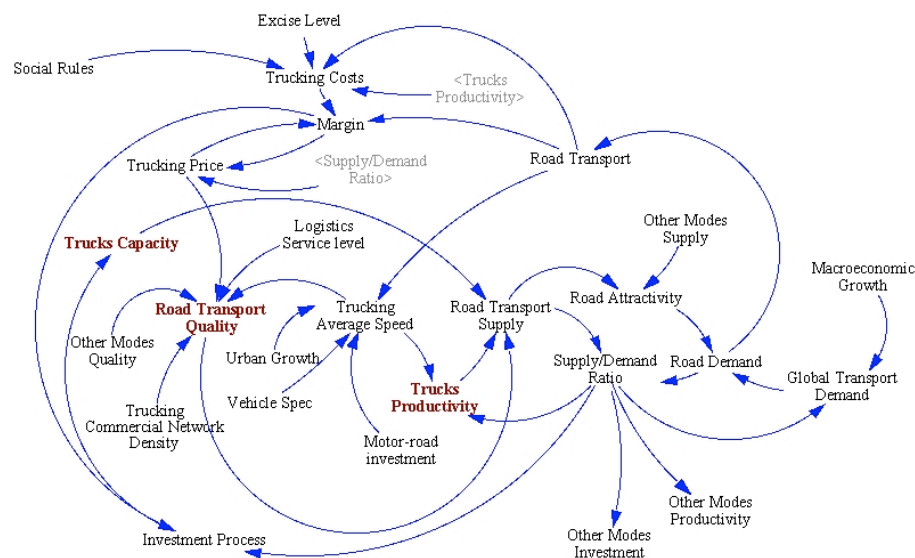


Figure 1. Simplified causal diagram of the model

The model contains one more sector corresponding to the economic environment, containing seven different freight and product types (agriculture, energy, chemicals, metallurgy, etc.)

Each transportation sector is defined by the following (figure 2) main variables, which are in fact whole sectors by themselves.

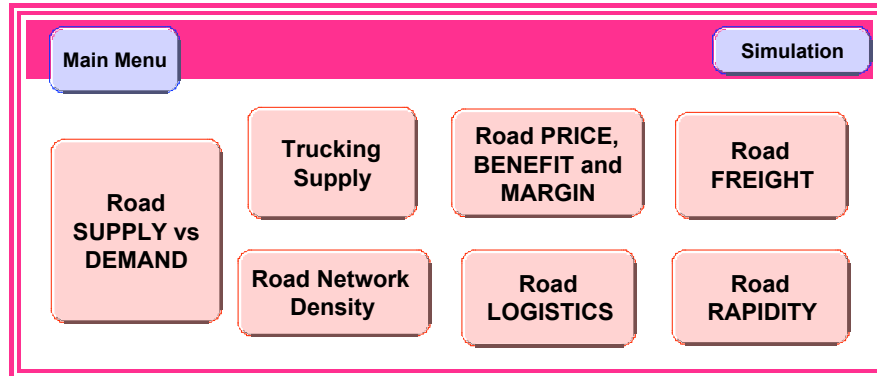


Figure 2. The Road's sector menu showing the different model blocks

As we say, we have considered that variables defining most freight transportation can be characterized by three main properties:

- How much (quantity),
- How efficient (productivity) ,
- How good (quality).

These triple defining properties apply to the amount of freight transported, to the commercial network density, to logistics as well as to rapidity. For example, road (as well as rail or river freight) is defined by available (and variable) Transportation Means, Freight Productivity (also a variable) and Transportation Quality.

The supply concept is very important for the transport industry. Transport cannot be stocked, and it is not possible to produce capacity “just in time”. What we need is an overcapacity, because shippers must believe that transport companies which they deal with are always able to bring the whole capacity into play. But in fact, supply is a combination of capacity, productivity and quality.

Capacity is basically a loading (usefull) “tonnage” of vehicules, barges, waggons, etc.

It is typically a stock depending on enterprises' behaviour. This level variable can be used with more or less efficiency, which we shall call “productivity” in the model. It represents the distance (kilometers) the available capacity can and does move per unit time. It depends on the supply/demand ratio on the market and some structural parameters.

That is to say that speed-regulations, quality of infrastructure, vehicle perfomance, allow a certain productivity , whether the market will use it or not.

The physical supply is the result of the multiplication of capacity by productivity, that is to say that you could produce a certain quantity of tons-kilometers.

Quality is a basic variable of transportation competition, and therefore explains the modal split.

We consider two types of quality parameters:

- The first category is the direct “quality” of the supply. What we describe is in terms of “conformity” (the right thing in the right place), and in terms of “no damage” to shipment. We consider that the percentage of damage and no conformity is equivalent to a decrease of the supply.

The following diagram gives a rough idea of the way we have modelled this triple concept of Means, Productivity and Quality, as applied, for example, to Road Transport.

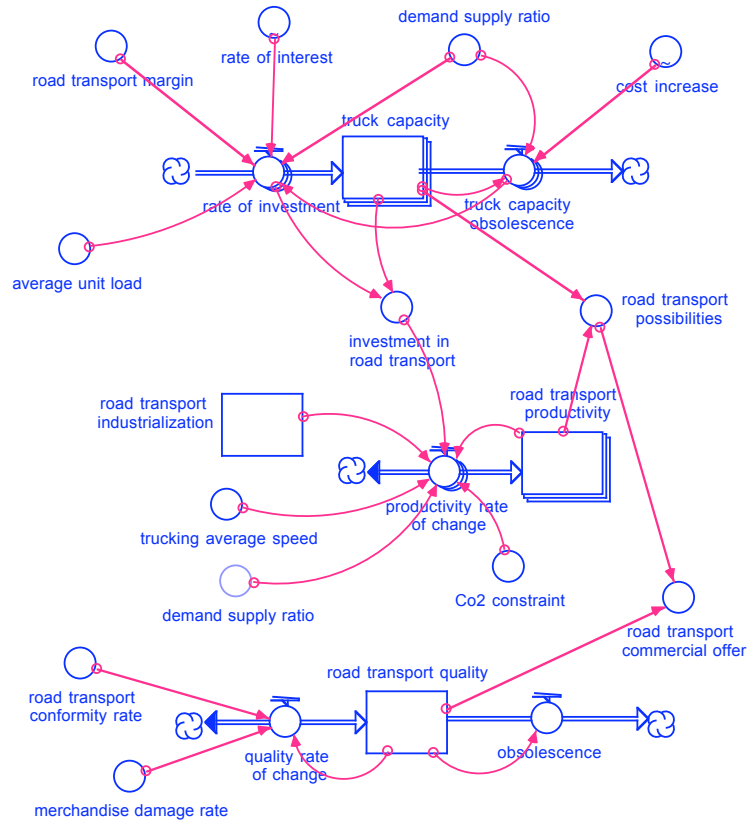


Figure 3. Road transport (supply) model (simplified)

– The second type of quality concept is more complex. It represents the way we compare rail, road or inland navigation. The problem is to combine different quality items such as price, price change, rapidity, network density, etc. in a globally attractive combination for each mode. Each quality index have not the same weight for the different modes.

For example, we can imagine that prices are very important for inland shipping, rapidity for road and rail, etc. The calibration of this part of the model is extremely important and combines statistical and experts analyses.

The “demand” sector attempts to deal with the diversity of exchanges. Transportation in France is the result of domestic trading, which means resident demand satisfied by French goods, importation goods driven by national demand, and French exportations driven by international (in fact mostly EU) demand.

The global level of exchange between other EU Nordic countries and the Iberic peninsula and Italy, also generates "transit" traffic through French territory.

Each basic economic sector has its own dynamics, both national and international, characterized by specific ratios (average distance, \$/ton ratio, international competitiveness, etc.)

The next causal graph shows the principles behind our sectorial forecasting process of demand.

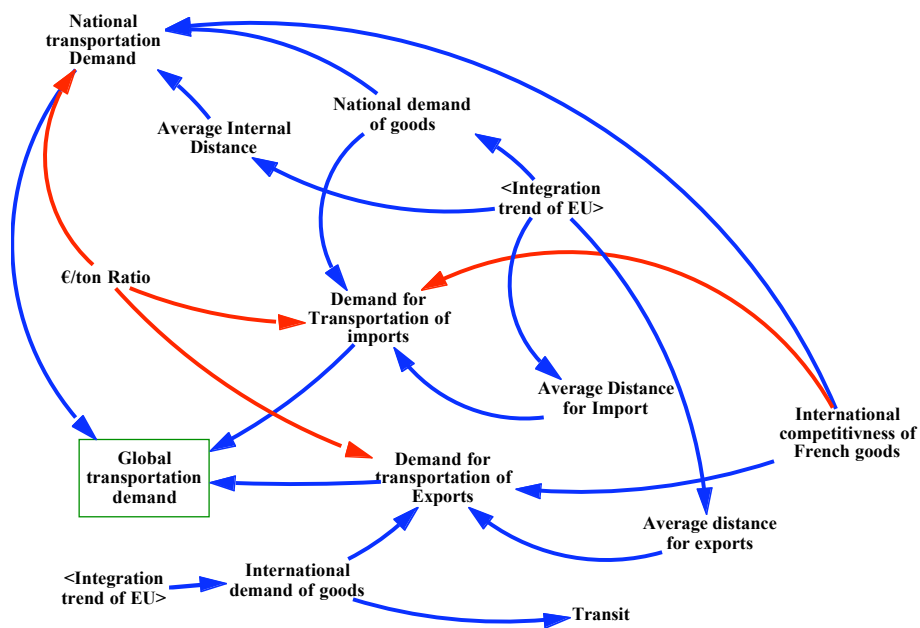


Figure 4. Demand Causal graph

In addition, we take into account the fact that the energy sector has links with the transportation sector – trucks use oil –, and that certain sectors have links among themselves. For example, fertilizers and chemicals have links with agriculture, and agriculture is linked with food industry.

The model can be used easily (Stella user friendly interface) by students or politicians (simplified inputs) or by professionals (differentiated inputs for each of the 7 economic sectors).

Three different ways to use the model have been defined:

- A “detailed” mode, where one can describe through graphical inputs a forecast over 20 years of most of the input variables of the model such as macroeconomic and freight sectors forecasts and transportation policy hypotheses;
- A “semi-detailed” mode, where the 7 freight sectors are endogenous;
- A “multiple-switch” mode, where the user has to choose between different ready-made macroeconomic packages and government actions.

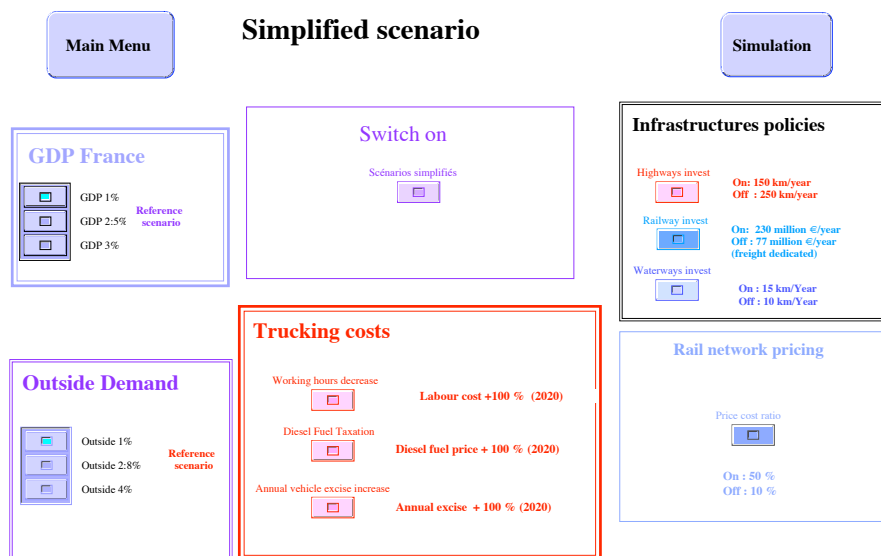


Figure 5. control panel for simplified use

The “simplified” mode is very useful to explain what is or is not possible in terms of transportation policy objectives.

For example, it is frequently suggested that a change in the road pricing policy, which would include accounting for external costs, would modify the modal split. Our simulations show that this assumption may be incorrect. An increase of the trucking labor cost (through lower allowed working hours) and a high level of excise tax, do not produce a significant change in the modal split.

On the other hand, with a new infrastructure policy, such as freight dedicated investment on railroad, and lower highway investment, the modal split could change significantly, although it would take some ten to twelve years.

The best practice policy is probably a narrow path for a government. For example, if the cost/price ratio for railroad infrastructure reaches an average of 50%

rather than 10%, all the benefits of the above mentioned policy package would be lost for rail.

This short description of the SimTrans model gives a general understanding of its mode of operation. Some remarks and criticisms can be made:

- The model does not take into account the geographical and technical variety of the transport offer. Thus, one does not take into account space disequilibria which weigh on this transport offer, nor the specificities of the equipments necessary to transport certain goods (cisterns, refrigerating vehicles, containers, buckets, etc.). To take into account such a variety would be impossible, since the decomposition of freight by type of products is not systematically related to the type of vehicle. In addition, the data available do not make it possible to suitably break down all the traffics on precise types of body. Taking these dimensions into account would lead to an enormous model...and a non-adjustable one for lack of numerical data. This last obstacle could be overcome when dealing with few variables which can be replaced by qualitative “dimensionless variable”, but it is totally unsurmountable when variables are numerous and of prime importance.

- Naturally, the equations of the model do not come directly from econometric studies.. The large amount of relations taken into account in the model does not allow for sufficient historical data which would validate meaningful causal links. Furthermore, such an approach would be meaningless. The basic principles of econometrics - statistical induction is meant to maximize the meaning of variance by minimizing the number of meaningful variables – are useless and inoperative when adjusting a System Dynamics model. The quality of the latter depends on the quality of the causal analysis, a fair data adjustment – often made difficult because of lack or poor quality of such available data – and in repeated simulations with careful analyses of sensitivity to changing parameters. Specialists of traditional modeling techniques do not always share this basic concept of simulation-based analysis. That is the reason why system dynamists like to show particularly the most counterintuitive results.

Classical models are based on the use of mathematical relations with a statistical adjustment of the effect of any variable on any other. For example, a rise in price has an influence – which can and must be measured (price elasticity) – on the consumption of a product. In a System Dynamics model, things are more complex. First of all, time: price changes with time, at a certain rate and a certain intensity. And this change will generate – still in time, but with various delays – behaviors, which can have contradictory influences on consumption. And these influences will depend on whatever situation we are in. In short, relations between variables change with time and interact with themselves

- And finally, a model such as SimTrans is at the same time too complex to be understood in depth by non prepared people, and on the other hand too simple. Learning through simulation requires a complex model, if only to be able to explore many scenarios and hypotheses. But to make these scenarios understood, requires

simple diagrams, with few variables and essential relations. We have tried to combine this triple approach to the modeling of a very complex problem:

- causal analysis,
- sophisticated quantitative or fuzzy qualitative modeling,
- causal diagramming of essential variables and relations necessary to the understanding of the likely causes of complex behavior of the model.

These are the steps used, in the following pages, to present SimTrans Co2.

3. Causal Loop Analysis

Causal diagrams generally constitute the first step in a coherent analysis of a problem, and they are generally drawn first, before formalizing and quantifying a model. Also, in a “System Thinking” type of approach, simple causal loops are drawn, and their behavior and possible effects are described.

In the present case, due in part to the complexity and size of the model, and the resulting difficulty in explaining both the model and its behavior modes, we were led to come back to a complete causal loop analysis after the model was built and used extensively. The purpose of such an *a posteriori* complex loop analysis is twofold:

- to help analyze and explain the various possible behavior modes of the system being analyzed,
- to show long term feedback loops whose existence and effect are very often not expected and/or ignored.

In practice, such a loop analysis is based on a causal diagram drawn after the complete model has been built, tested and extensively used. Hence both modelers and to a certain extent users of the SD model know what really lies behind each causal variable and behind the causal arrows. We have experienced that during such *a posteriori* causal analyses, frequent switches from causal to dynamic models are very useful and help understand both the general structure and the details of a model.

In a real model, even if limiting oneself to a one page causal diagram such as the Causal Diagram of SimTrans Co2, loops easily number by scores or hundreds. For the purpose of analysis and understanding, there are three types of loops:

- Short loops (meaning that few variables are involved) which may have short or long term effects. Such loops are easy to explain, most often they are self-evident and experts as well as model users do not feel that they get much benefit from their description and analysis, rather that they are wasting their time.
- Long loops (involving at least 10 variables or more): some may have short term effects, others being rather long term, but in both cases these dynamics do correspond to expected behavior. The analysis of such loops is useful because the

corresponding effects are generally a mix of many loop behavior types, and a loop by loop decomposition helps understand what was possibly expected, hoped for or forecast through intuition, but was too complex to be rationally and simply analyzed and explained.

– Long loops (in general very long loops), with long term and sometimes counterintuitive effects, but whose existence is ill perceived, or ignored or forgotten. To show and analyze these loops one by one, showing also how they can combine effects, has proven to be very effective in the process of convincing users, experts and clients, that their problem is in effect a complex one and that long term consequences of decisions are, and can be forecast and taken into account.

For example, here is the causal diagram drawn after SimTrans Co2 was used:

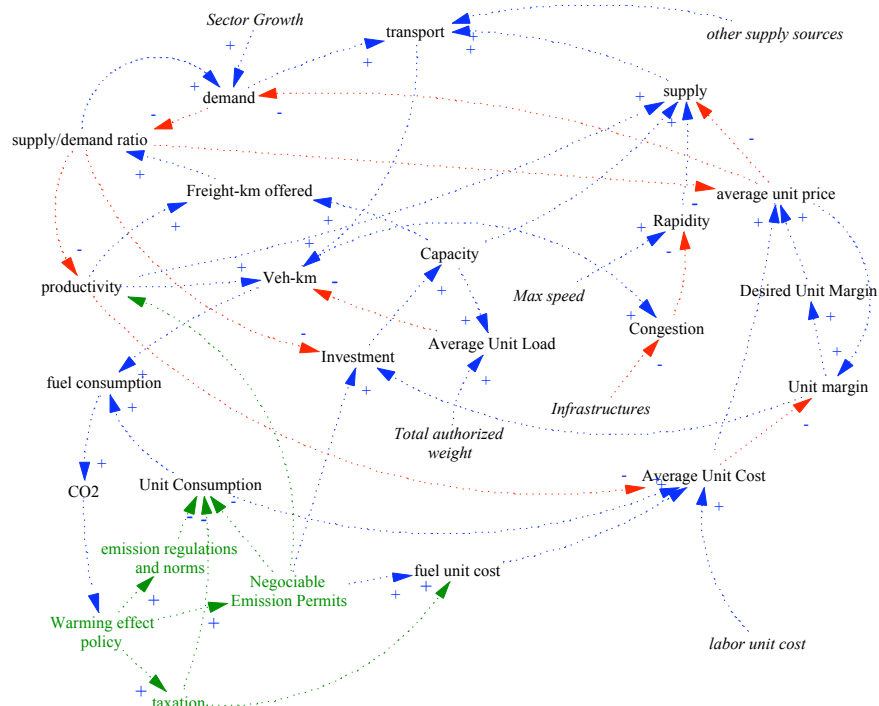


Figure 6. A general Causal Diagram of SimTrans Co2

Applying the Vensim loop analysis, we find that there are some 200 loops within this causal model (for example, 159 of them go through the Fuel Consumption variable). As already mentioned most of the short term loops are easy to analyze, and correspond to one's knowledge, experience or intuition. But many loops, mostly those with long term dynamics, are not evident, are seldom taken into account or are

even ignored. These long term loops are the most « interesting » and are of utmost importance, since their effect occurs late in the whole process, when the initial cause is long-forgotten and not traceable any more. Although unavoidable and often troublesome, these long-term effects seem hard to understand or at least unclear, since one cannot come back to the initial cause of disturbance. As can be noticed, the generally ignored long term loops are not only long term in the dynamic behavior sense, they are also long, that is they involve many interconnected variables, which is one reason for the difficulty of dealing with them.

Let us give some examples:

The loop shown in heavy lines in the following diagram, is a relatively long term one, but it is evident, and graphically short. It says that a lot of transportation leads to traffic congestion, hence to a slow down of traffic as well as transportation supply a stabilizing loop. But that loop is well known, felt every workday, and showing it to experts of the field is of no use, and is even counter effective.

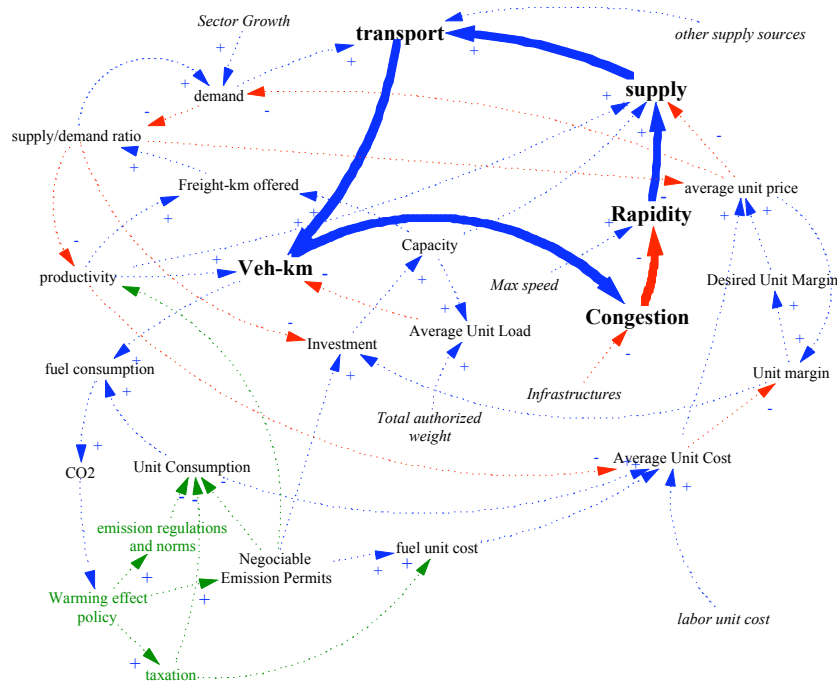


Figure 7. A long-term « evident » feedback loop

The following loop is less evident because it involves many variables, hence it is less intuitive and requires the help of tools such as the Vensim loop analysis one. Nevertheless, its potential effect is not surprising since it corresponds to the hoped for effect of taxation on emissions of Co2.

Taxation leads to increased fuel prices, which tends to increase freight transportation prices, and decrease the supply of transportation, hence the amount of vehicle-km and the total consumption of fuel. The corresponding decrease of emission of Co2 is the desired effect of that negative loop, but it would also tend, if one is not careful about this effect, to weaken the taxation rules.

This could be a relatively fast (one or two year time constant) feedback loop, which does correspond to the expected effects.

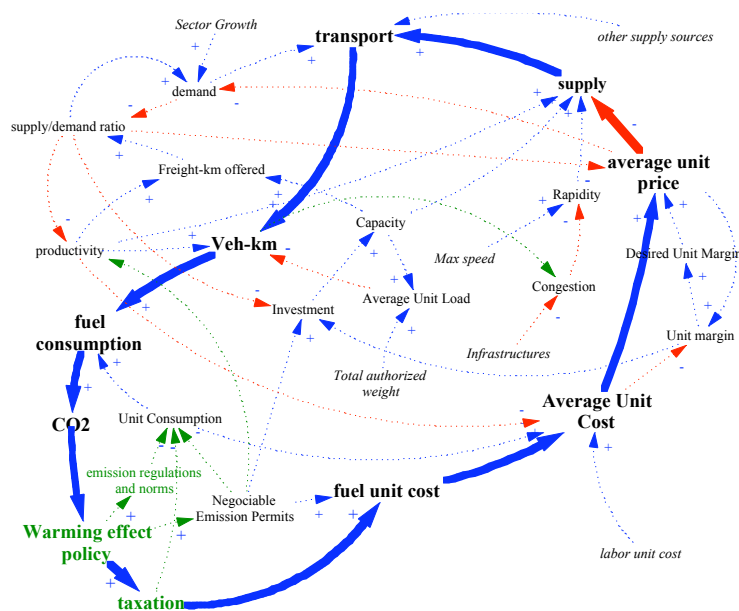


Figure 8. A relatively fast and hoped-for feedback loop

On the other hand, the next diagram shows a very similar loop whose effect would be contrary to the one shown in the previous loop. Taxation would lead to an increase of fuel unit cost, but also, a little later and progressively, to a reduction in unit consumption, hence to a decrease of average unit cost of transportation. This positive feedback loop may, in the long run, compensate the previous stabilizing loop. The right questions to be asked would be:

- what would be the respective increase of unit cost and decrease of unit consumption?,
- what are the approximate time constants of each link, in particular how soon would the unit consumption start to decrease?

Only experts of the field can possibly answer such questions (trying to do so results generally in an increase of their own expertise), but the simulation model itself constitutes a means of checking the reliability of the new data.

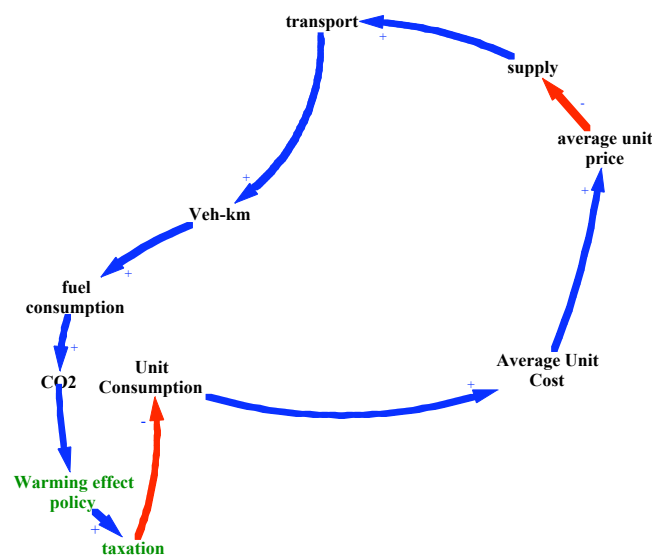


Figure 9. A slower positive feedback loop, potentially counteracting the previous one

Let us show a few more long term, sometimes counterintuitive, loops which often appear in counteracting pairs.

The following diagram shows such a pair of counteracting loops, which are both long term since they involve slow moving Investment, Capacity and Average Unit Load variables. Creating Emission Permits may thus have a long term favorable effect on the emission of Co2, but also, and first of all, a negative consequence on that same Co2 emission variable.

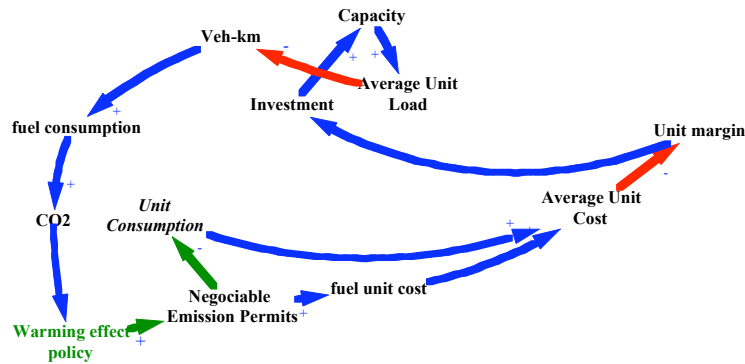


Figure 10. Two long term counteracting loops

The following two loops are potentially favorable to a decrease in Co2 emission, but the effect can be particularly slow in the first case, and even counterintuitive in the second case (below) since it involves both offer and demand, but through an indirect productivity effect (if demand decreases, this will have a negative effect on productivity, hence on supply, but only after a long delay).

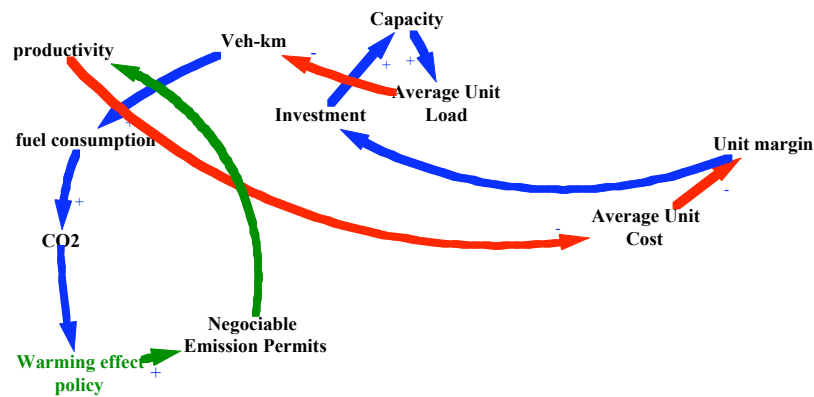


Figure 11. Potentially favourable loop (#1)

Making such loops explicit does not solve problems, but it helps analyze and understand long term hidden, sometimes counterintuitive effects. But in order for such an analysis to be really useful, these loops and the corresponding causal

diagrams must be based on a well tested and thoroughly understood complex dynamic model.

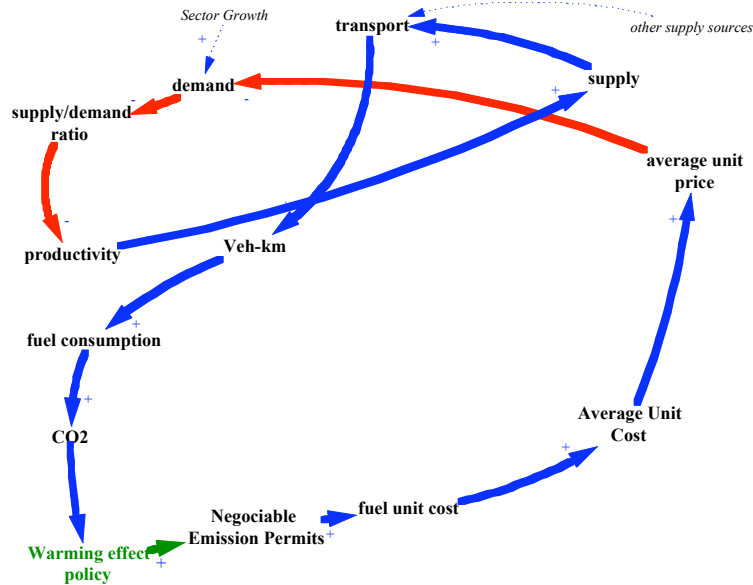


Figure 12. Potentially favourable loop (#2)

In a way, such a combination could be called an « *ex-post* Complex System Thinking ».

4. Dynamic simulation of emission permit policies

But systems dynamics is above all a powerful time simulation tool. The two models SimTrans (which can forecast the evolution of freight transport in France) and SimTransCo2 (which analyzes the potential effects of negotiable permits for greenhouse gas emission) have as their first aim, to give quantified forecasts.

The SimTrans model has been adapted and used to try to answer the following question, asked by the French Energy and Environment Governmental Agency (ADEME) : what does the Kyoto proposed Emission credit policy do to the freight transportation sector, and conversely, which policy concerning this sector is likely to have an effect on the emission of CO₂?

In March 2000, the Commission launched the European Climate Change Programme (ECCP) to prepare additional policies to ensure that the EU achieves the

8% cut in Co2 emission by 2008-2012, as compared with the level of 1990, a cut to which it is committed under the Kyoto Protocol. The latest data show that Co2 emissions are increasing rather than decreasing, and that the 8% reduction goal will not be met if no additional measures are taken. And transportation emissions are growing in OU at even a higher rate.

As stated in the Commission's November 2000 Green Paper on security of supply, in 1998 energy consumption in the transportation sector was to blame for 28% of emissions of CO₂, the leading greenhouse gas. According to the latest estimates, if nothing is done to reverse the traffic growth trend, CO₂ emissions from transportation can be expected to increase by around 50% to reach 1 113 billion tonnes in 2010, compared with the 739 million tonnes recorded in 1990. Once again, road transportation is the main culprit since it alone accounts for 84% of the CO₂ emissions attributable to transportation. However, internal combustion engines are notorious for their low energy efficiency, mainly because only part of the combustion power serves to move the vehicle (European, 2001).

According to the ECCP, one strong policy strategy would consist in a internal EU greenhouse gas emission trading scheme. It is this proposed strategy that we were asked to analyze through the use of an adapted version of SimTrans.

A simplified causal diagram is shown below (figure 13), which will be combined with the causal diagram of SimTrans.

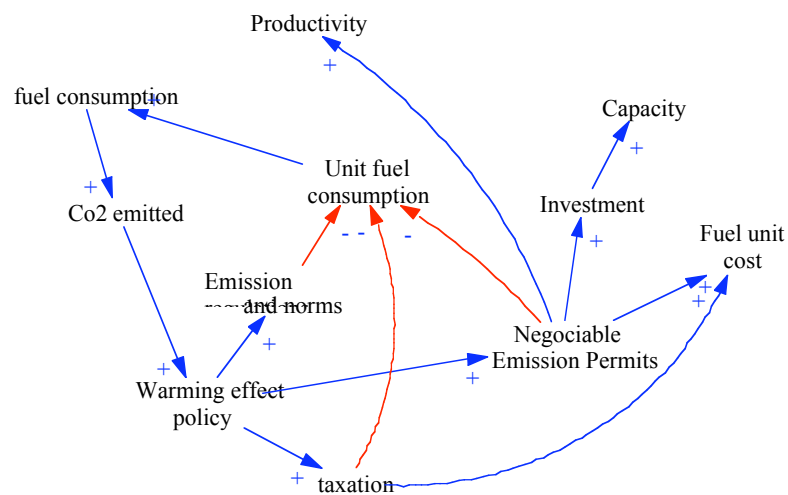


Figure 14. Simplified causal diagram of Co2 and emission permit addition to SimTrans

This portion of the SimTrans Co2 model contains two parts:

- a permit allocation sector, which models various policies for allocation of permits, their use, their sale or acquisition (buying). The allocation itself can be free (or not), periodic, recursive, diminishing, etc.
- an Energy consumption and saving sector which shows possible links between permit allocation and use, and change in driving habits, capacity usage and productivity, etc., such links being either voluntary or imposed on the freight transportation sector.

The French national program against climate change projected a carbon tax which represents a fuel excise of 0.053 Euros *per* liter. Using this model has shown that a permit trading scheme, based on such a value of permit price, has no real impact on road transportation and modal split. Neither does the impact change drastically with higher Co2 permit price levels.

Curve 2 in the following graph shows such a profile – with an emission credit price of 0.9 Euros/liter – which is not very different from the passed trend (curve 1). Note that this “price” is not an average one for the trucking industry, but a marginal fuel cost. This marginal cost “appears” when the level of Co2 emission becomes higher than what is allowed. Note also that such a value, which may appear high, is not very far from the changes of the average diesel fuel price witnessed during the last two years. We know that economists are anticipating an impact on producer behavior when average costs are significantly lower than marginal ones. They consider that marginal costs drive the pricing behaviour. But, in the market competition, other variables are very important. There are many feedback loops (see part 4 – Causal Loop Analysis) which potentially can and are likely to counteract the “initial price effect”.

For example, a difference between marginal and average fuel costs will accelerate the willingness to buy new and more efficient trucks. It will generate a new interest for drivers training, and probably increase the capacity of new trucks and put the emphasis on the management of capacity utilization. After some time, our model shows that these “induced, so-called secondary effects” could become very important and may more or less rapidly and completely compensate the initial effect of increasing the marginal cost above the average one.

On the other hand, we have tested the effect of a policy package with excises, labor rules, investment expenses, etc. Curve 3 indicates that such a global policy “package”, which includes Co2 emission trading plus the following policies: decrease of working hours, diesel fuel taxation and railroad infrastructure investments (*cf.* presentation of “Simplified mode control panel”), could be efficient.

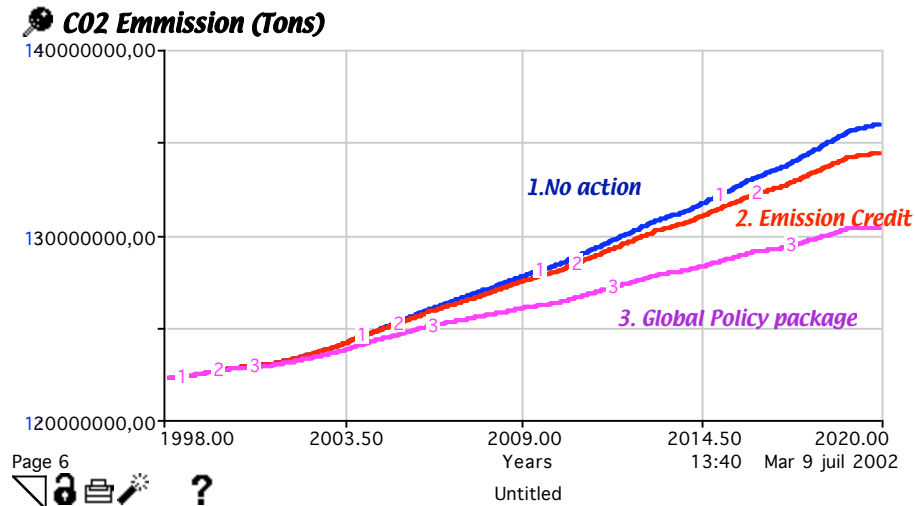


Figure 14. Comparative graphs of Co2 emission

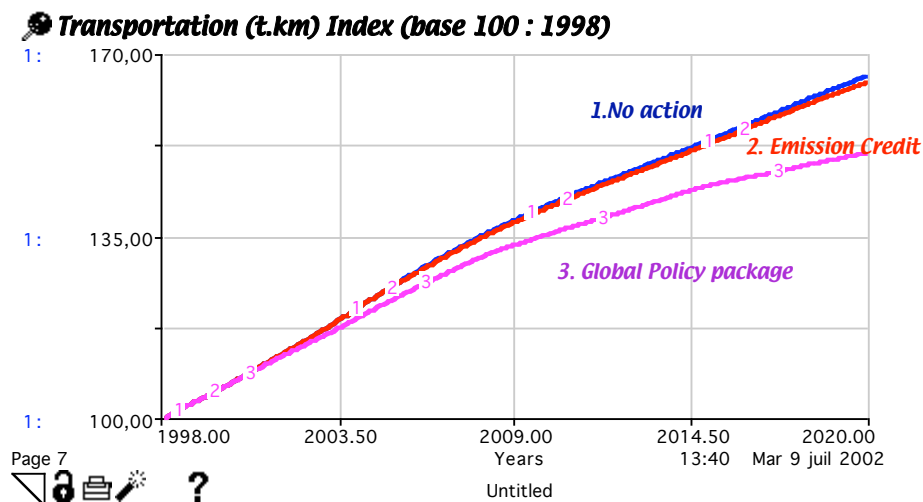


Figure 15. Comparative graphs of transportation

It will be of no surprise to find that a policy applying a strict quota on Co2, specific to the road transportation sector, plus an internal emission trading policy, would be efficient, but might probably generate a global deflation impact on economy. Such a policy would be drastic: trucks would not be allowed to produce

more than a maximum level of Co2. It is likely that the industry could imagine strategies to save energy, use bigger trucks, increase the allowed maximum load, replace old engines, etc. But when reaching the maximum allowed Co2 level, there would be no other solution than to cancel whatever remaining freight transportation was yet to be done, which really means decreasing the level of supply. Such a scenario (see next graph) would lead to a “transportation crisis”: for the first time since the XIXth century, the transportation system would not be able to satisfy demand. This could lead to a decrease of the GDP, a scenario which could be analyzed only by means of a totally different model, which would have to take into account social and political repercussions, among others. Such a model is yet to be built.

Co2 Emission (tons)

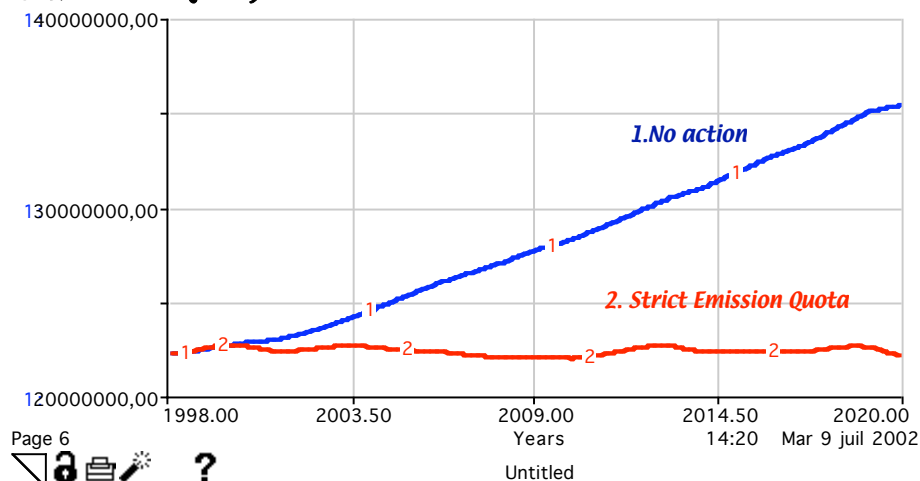


Figure 16. Comparative graphs of Co2 emission

These approaches are those which correspond to the “decoupling” mobility (human beings and goods) from economic growth thesis. This is not what is done in SimTrans or SimTransCo2. Transport which have not been carried out, simply weigh on the future demand satisfaction (carryforward). But isn’t this problem of “global looping” the most interesting although extraordinarily complex ?

It must combine:

- The Analysis and modelling of critical system dynamics, with the purpose of implementing a strategic simulator for the European Transport of Goods, and providing a decision support tool in terms of backcasting where significant changes in the intermodality structure are demanded by sustainable solutions;
- A Corridor studies, including traffic and network planning and spatial modelling, with the purpose of providing a geographical analysis of the system,

particularly important for international traffic, and cohesion on the European continent;

– And a Global economic impact analysis of transport policies in order to build the feedbacks between transport systems and economic system.

Transportation (t.km) Index (base 100 : 1998)

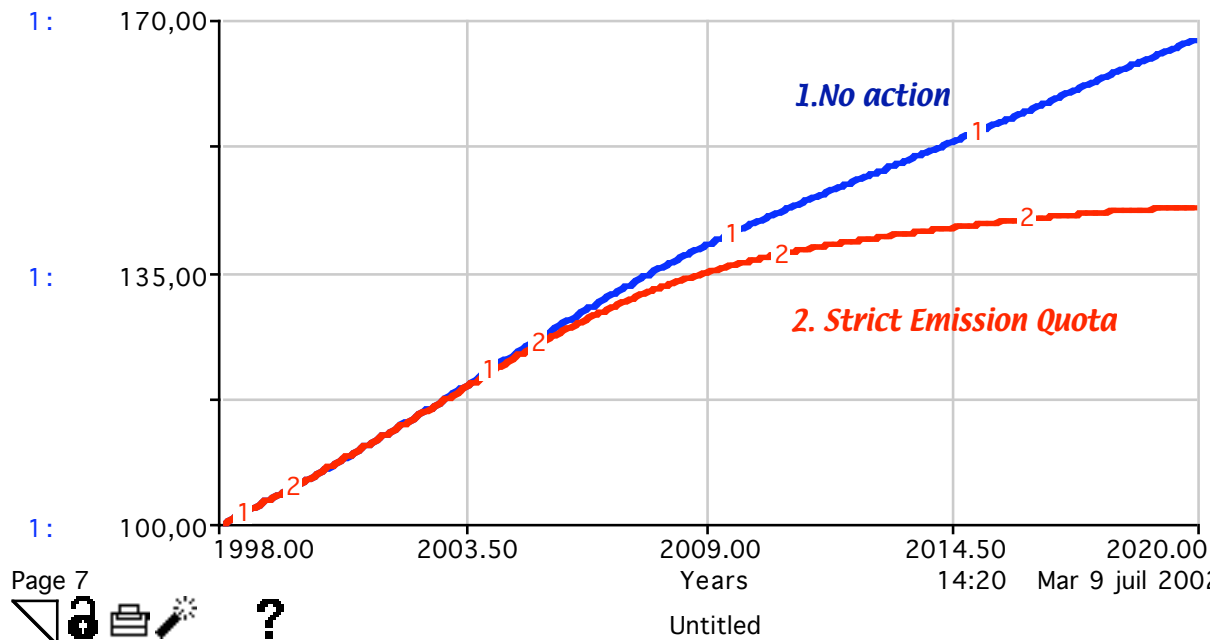


Figure 17. Comparative graphs of total transportation index

This first system dynamics exercise (SimTrans & SimTransCo2) dealing with transportation freight economics in France generated a renewal (or rebirth) of System dynamics in transport sector. New networks, new projects are going to explore this “old new” way to analyse this so called complex transportation system. Probably, the “sustainable development thematic” will be a good support for this fundamental renewal of forecasting approach.

By another way our model shows very clearly that the freight transportation system needs complex policy package – combining investment, regulations etc. – to begin to deal with sustainable development objectives.

References

- European, “European transport policy for 2010 : time to decide”, White Paper of the European Commission, 2001.
- Ecmt, “Trends In The Transport Sector 1970-2000”, *European Conference of Ministers of Transport*, OECD/Ecmt, 2002.
- Houée M., De Jong G., Jiang F., “Modélisation du choix modal en transport de marchandises à partir de la combinaison de données en préférences révélées et déclarées”, Notes de synthèses du SES, 2002.
- Kbs, “Simtrans CO2”, 2002.
- Salini P., Chatard D., Harcouet C., Sae, “Pretram”, 1981, 1982 and 1983.
- Salini P., Karsky M., « Modèle de simulation dynamique des transports de marchandises : Un outil dédié à la prospective », Transports, mai-juin 2000.
- Schade W., “ASTRA” (Assesment of Transport Strategies) final report, IWW, Universität Karlsruhe, 2000, <http://www.iww.uni-karlsruhe.de/ASTRA/deliverables.html>.
- Sterman J., McGraw-Hill I., *Business Dynamics*, 2000.
- TERM, Transport and environment reporting mechanism, 2002 Report, European Environment Agency, 2002, <http://themes.eea.eu.int/>
- Think-Up, “Segmentation of Freight Markets”, (Think-Up is a thematic network project), *European Commission's 5th Framework Programme for Research and Development*, 2001, <http://www.netr.fr/think-up/uk/reports/>.